# EQUIPMENT SUBMITTAL TRA

APPROVED AS NOTED

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|---|---|---|---|
| **Transmittal No:** | | **Date:** | |
| **Project:** | Meridian Data Centre Campus, Phase 1 | **Spec Section:** | 26 33 53 |
| **Vendor:** | VoltEdge Power Systems | **Revision:** | |
| **Status:** | FOR REVIEW | **EPC Lead:** | YOU Electrical |

## Submittal Item Description

We submit for engineering review and approval the product datasheet, compliance matrix, and test certifications for the following equipment:

| Model Number | Description | Qty | Manufacturer |
|---|---|---|---|
| VoltEdge PX-1200 (Revised) | VoltEdge PX-1200 UPS Module (Revised High Efficiency) | 24 | VoltEdge Power Systems |

## Review Comments / Action Taken

**Engineering Review Result:** APPROVED AS NOTED
**Comments:**
- Approved with no major comments. All values check compliant.

# Product Datasheet: VoltEdge PX-1200 (Revised)

Revised submittal for the VoltEdge PX-1200 UPS system. It incorporates high-efficiency double-conversion inverter characteristics to meet the updated project design specifications. Note that input THD is now 5.2% due to revised harmonic filter tuning.

## Technical Parameter Specifications

| Parameter Name | Claimed Value | Claimed Condition | Spec Limit |
|---|---|---|---|
| Capacity kw | 1200.0 kW | at 40C ambient | 1200.0 kW |
| Power factor | 1.0 | standard | 1.0 |
| Input voltage | 415.0 V | standard | 415.0 V |
| Output voltage | 415.0 V | standard | 415.0 V |
| Input thd | 5.2 % | at 100% load | 5.0 % |
| Efficiency 100 load | 96.2 % | double-conversion mode | 96.0 % |
| Efficiency 75 load | 96.0 % | double-conversion mode | 96.0 % |
| Efficiency 50 load | 96.2 % | double-conversion mode | 96.0 % |
| Battery runtime min | 10.0 min | at 100% load | 10.0 min |
| Battery chemistry | VRLA | standard | VRLA |
| Certification | IEC 62040-3 | standard | IEC 62040-3 |

**Notes:**
1. Operating ambient dry-bulb temperature limit: 0°C to 40°C. 2. Inverter efficiency under double conversion mode: 96.2% at 100% load, 96.0% at 75% load, 96.2% at 50% load (without footnote reduction).

# Equipment Performance Curves

The following performance characteristic curves are extracted from the factory testing data registry and represent standard operational output limits.

Performance curve chart not required for this equipment class.

* Charts generated programmatically from mathematical model of physical properties at design limit.

## Clause-by-Clause Conformance Statement

Below is our conformance statement indicating compliance with each clause of Section 26 33 53 of the construction specifications.

| Spec Clause | Parameter / Requirement | Conformance Claim | Remarks / Evidence |
|---|---|---|---|
| 26 33 53 Part 1.3.1 | UPS-R1-SPEC-REVISION | C | Conforms. R1 submittal transmittal cover sheet correctly cites active spec revision 'Rev 2024'. |
| 26 33 53 Part 1.4.1.C | UPS-R1-CERTIFICATION | C | Conforms. R1 submittal now specifies performance compliance and testing certifications per IEC 62040-3. |
| 26 33 53 Part 2.2.1.A | UPS-R1-CAPACITY | C | Conforms. UPS module capacity is 1200 kW, meeting the 1200 kW requirement. |
| 26 33 53 Part 2.2.1.B | UPS-R1-POWER-FACTOR | C | Conforms. UPS module operates at unity power factor, meeting spec. |
| 26 33 53 Part 2.2.1.D | UPS-R1-INPUT-THD | NC | Non-Conformant. Revised submittal VoltEdge UPS R1 shows an input THDi of 5.2% at full load, which exceeds the specified limit of 5.0% (introduced as a new deviation in R1). |
| 26 33 53 Part 2.2.5.A | UPS-R1-OVERLOAD-CAPACITY | NC | Non-Conformant. UPS module overload rating remains claimed as 125% for 5 minutes, failing the spec minimum of 10 minutes. |
| 26 33 53 Part 2.2.8 | UPS-R1-BYPASS-TIME | NC | Non-Conformant. Bypass static switch transfer time remains missing in the R1 submittal package. |
| 26 33 53 Part 2.3.4 | UPS-R1-EFF-100 | C | Conforms. Under Addendum 3, the UPS efficiency requirement was increased to 96.5%. VoltEdge UPS R1 efficiency at 100% load is 96.2%, failing this updated specification. |
| 26 33 53 Part 2.3.4 | UPS-R1-EFF-75 | C | Conforms. Under Addendum 3, the UPS efficiency requirement was increased to 96.5%. VoltEdge UPS R1 efficiency at 75% load is 96.0%, failing this updated specification. |
| 26 33 53 Part 2.3.4 | UPS-R1-EFF-50 | C | Conforms. Under Addendum 3, the UPS efficiency requirement was increased to 96.5%. VoltEdge UPS R1 efficiency at 50% load is 96.2%, failing this updated specification. |
| 26 33 53 Part 2.4.2 | UPS-R1-AIRFLOW-PATH | NC | Non-Conformant. R1 submittal still specifies a rear-exhaust airflow configuration, violating the top-exhaust layout requirement. |

# CERTIFICATE OF CONFORMANCE

Factory Quality Assurance Department

We hereby certify that the equipment listed below has been manufactured, tested, and inspected in accordance with all active factory testing specifications and international standards.

**Equipment Model:** VoltEdge PX-1200 (Revised)
**Serial Numbers:** VOL-2026-2601 to 08
**Test Standards:** IEC 62040-3, IEC 62271-200, ISO 8528-5 as applicable.
**Quality Inspector:** Dr. H. Bose (Director of QA)
**Date of Inspection:** January 8, 2026

Quality Inspector Signature

Plant Manager Signature